Assignment 2

Problems:

1. Negate the following statement: For every real number $\epsilon > 0$ there exists a positive integer k such that for all positive integers n, it is the case that $|a_n - k^2| < \epsilon$.

2. Let $f : X \to Y$, $A, B \subseteq X$, and $C, D \subseteq Y$. Show that f^{-1} preserves inclusions, unions, and intersections.

   [Hint: Recall that $f(A) \triangleq \{f(x) | x \in A\}$ and $f^{-1}(C) \triangleq \{x \in X | f(x) \in C\}$.]

   (a) Show that $C \subseteq D \implies f^{-1}(C) \subseteq f^{-1}(D)$.

   (b) Show that $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.

   (c) Show that $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.

   Show that f preserves inclusions and unions, but not intersections.

   (d) Show that $A \subseteq B \implies f(A) \subseteq f(B)$.

   (e) Show that $f(A \cup B) = f(A) \cup f(B)$.

   (f) Show that $f(A \cap B) \subseteq f(A) \cap f(B)$. Give an example where equality does not occur.

3. Let $\underline{x} = (x_1, \ldots, x_n), \underline{y} = (y_1, \ldots, y_n) \in \mathbb{R}^n$ and consider the function ρ given by

$$\rho\left(\underline{x}, \underline{y}\right) = \max\left\{|x_1 - y_1|, \ldots, |x_n - y_n|\right\}.$$

   Show that ρ is a metric.

4. Suppose $a \in B_d(x, \epsilon)$ with $\epsilon > 0$. Find an explicit $\delta > 0$ such that the open ball $B_d(a, \delta)$ centered at a is contained in $B_d(x, \epsilon)$.

5. This problem outlines a proof that the Euclidean distance d on $\mathbb{R}^n$ is a metric, as follows: If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $c \in \mathbb{R}$, define

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \ldots, x_n + y_n) \qquad c\mathbf{x} = (cx_1, \ldots, cx_n),$$
$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + \cdots + x_n y_n \qquad \|\mathbf{x}\| = (\mathbf{x} \cdot \mathbf{x})^{1/2}.$$

   (a) Show that $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = (\mathbf{x} \cdot \mathbf{y}) + (\mathbf{x} \cdot \mathbf{z})$.

   (b) Show that $|\mathbf{x} \cdot \mathbf{y}| \le \|\mathbf{x}\| \|\mathbf{y}\|$. [Hint: If $\mathbf{x}, \mathbf{y} \neq 0$, let $a = 1/\|\mathbf{x}\|$ and $b = 1/\|\mathbf{y}\|$, and use the fact that $\|a\mathbf{x} \pm b\mathbf{y}\| \ge 0$.]

   (c) Show that $\|\mathbf{x} + \mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$. [Hint: Compute $(\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y})$ and apply previous result.]

   (d) Verify that the Euclidean distance $d(\mathbf{x}, \mathbf{y})$ is a metric.

6. Define $f_n : [0, 1] \to \mathbb{R}$ by the equation $f_n(x) = x^n$. Show that the sequence $\{f_n(x)\}$ converges for each $x \in [0, 1]$, but that the sequence $\{f_n\}$ does not converge uniformly. Recall that uniform convergence to f on $[a, b]$ implies that, for any $\epsilon > 0$, there exists an N such that $|f_n(t) - f(t)| < \epsilon$ for all $n > N$ and all $t \in [a, b]$.

Optional Problems:

1. Negate the following statement: There exists an integer Q such that for all real numbers $x > 0$, there exists a positive integer k such that $\ln(Q - x) > 5$ and that if $x \leq k$ then Q is cacophonous. (The last term used in this exercise is meaningless.)